

Sandhya Nunemunthala

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PROFESSIONAL SUMMARY:

- **11+ Years** of experience in designing, developing, and maintaining **AWS (Amazon Web Services), GCP (Google Cloud Platform) and Microsoft Azure**
- Certified in both AWS and GCP
 - ⇒ **AWS Solution Architect Professional** – [Certification Link](#)
 - ⇒ **Google Cloud Certified Professional Data Engineer** – [Certification Link](#)
- Designed and developed POC Solutions for **Gen AI** Data Pipeline using **AWS Cloud** and **GCP Cloud**.
- Designed and developed multi-tenant data applications supporting **Data Warehouse, Data Lake** and **Data Mart** in **AWS** and **GCP Cloud**
 - ⇒ In AWS Cloud I have used services like: **Redshift, S3, RDS (Oracle, MySQL, MSSQL), Glue (Jobs, Connections, Notebooks, Crawlers & Catalog), EMR Cluster, Ec2, Step Function & Airflow, SNS Topic & SQS, Lambda (Functions & Layers), IAM (Roles & Policies)**, etc.,
 - ⇒ In GCP Cloud I have used services like: **BigQuery, Cloud Storage, Cloud SQL (Oracle, MySQL), Data Flow, Data Proc, Compute Engine, Cloud Composer, Pubsub, Cloud Function, IAM (Roles & Service Accounts)**, ect.,
- Hands-on experience designing and implementing **Generative AI pipelines** and **LLM-based solutions** across **AWS** and **GCP** cloud ecosystems to support **chatbots, agentic workflows, and context-aware automation**.
- Designed and developed **Proof-of-Concepts (POCs)** for **Gen AI** applications using **Amazon Bedrock, SageMaker JumpStart, Titan Embeddings, and Vertex AI** with **Gemini models** to explore use cases in customer service, search augmentation, and document intelligence.
- Created intelligent agents and workflow automations using **Bedrock Agents, Amazon Lex, Comprehend**, and **AWS Step Functions** to deliver context-aware, goal-driven task orchestration across cloud-native services.
- Designed and developed highly scalable data applications for Machine Learning, Data Science and Generative AI Solutions to support real-time and batch use case in **AWS** and **GCP Cloud**
 - ⇒ In AWS Cloud I have used services like: **Glue, Lambda, OpenSearch, SageMaker, S3, DynamoDB**,
 - ⇒ In GCP Cloud I have used services like: **Cloud Run, Cloud Function, AlloyDB, Vertex AI, Document AI, Gemini, BigQuery**
- Designed and developed **Model Monitoring** solutions to validate model performance and **model Metrix** in **AWS Cloud**
- Designed and developed feature store for **Machine Learning** and **Data Science** use cases in **AWS Cloud**
- Designed and developed real-time and batch reports and Dashboards in both **AWS** and **GCP** cloud
 - ⇒ In AWS Cloud I have used services like: **S3, Lambda, Glue, SNS, Cloud Watch Event StepFunction and QuickSight**
 - ⇒ In GCP Cloud I have used services like: **Cloud Storage, Cloud Function, Pubsub, Cloud Scheduler, Workflow, Cloud Composer, BigQuery** and **Power BI**
- Extensive work experience in Implemented **continuous integration and continuous deployment (CI/CD)** through Bit Bucket and Jenkins using **UCD (UrbanCode Deploy)** also worked extensively using Gitlab and **Terraform**.
- Strong expertise in building scalable ELT/ETL pipelines using Databricks, Apache Spark, Airflow, AWS Glue, and cloud-native services.

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- Experienced in implementing data governance, observability, monitoring, lineage, and IAM-based access control in regulated environments.
- Extensive experience in writing **Infrastructure as Code (IaaC)** in **Terraform, AWS CloudFormation**
- Designed and developed purpose-built layers using **AWS** services like **RDS (Oracle, MySQL), Redshift, DMS (Data Migration Service), S3, Cloud Watch alarms.**
- Specialized in end-to-end **Databricks Lakehouse implementations**, including Delta Lake architecture, Spark optimization, and multi-tenant data platform design.
- Developed complex mappings and loaded data from various sources into the **Data Warehouse, using different transformations/stages like Joiner, Transformer, Aggregator, Update Strategy, Rank, Lookup, Filter, Sorter, Source Qualifier, Stored Procedure transformation, etc.**
- Extensive working experience in Apache Stack: **HDFS, Hadoop, Hive, MapReduce, Spark, Kafka, Zookeeper, Airflow, Flume, Super Set**
- Strong expertise in **Python and SQL with distributed data processing frameworks such as Spark, Kafka, and Airflow**, building high-volume ETL/ELT pipelines and event-driven architectures.
- Experienced in **lake house architectures, dimensional data modelling, and GenAI/RAG pipelines**, delivering governed, high-quality data platforms that support advanced analytics and AI-driven applications.
- Scheduled and Orchestrated data applications for both real-time event triggering jobs and batch in both **AWS** and **GCP Cloud**
 - ⇒ In AWS Cloud I have used services like: **Cloud Watch Event, Lambda, SNS Topic, Airflow, Step Function, S3, Glue (Jobs, Connections, Notebooks, Crawlers & Catalog), Redshift, Cloud Watch logs**
 - ⇒ In GCP Cloud I have used services like: **Cloud Scheduler, Workflow, Cloud Composer, Pub-Sub, DataFlow, BigQuery, Cloud SQL**
- Migrated legacy data applications from **Hadoop** ecosystem to **AWS Cloud** and **GCP Cloud**.
- Used **QlikView, Kibana, Tableau** and **Apache superset** for visualization.

PROFESSIONAL EXPERIENCE:

Prospect Intel, Albany, NY

Jul 2025- Till Date

Role: Senior AI ML Engineer

Roles & Responsibilities:

- Led end-to-end delivery of **AI-powered applications**, working closely with clients to understand business needs and convert them into reliable, cloud-native solutions.
- Developed and operationalized a **centralized AI Prompt Framework** to standardize prompt engineering across features and maximize LLM accuracy, consistency, and reusability.
- Built scalable AI event pipelines (real-time, scheduled, batch) using **Cloud Run, Vertex AI, Gemini (Flash & Pro), Google Workflows**, and **Cloud Scheduler**, ensuring efficient orchestration and automation.

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- **Architected and delivered an Agentic AI MVP** leveraging **Google Gemini** and the **Agent Development Kit (ADK)** to enable autonomous reasoning, task execution, and intelligent decision-making across application workflows.
- Designed **low-latency REST APIs** with **FastAPI** and **Uvicorn** to aggregate data from internal databases and external data providers, enrich it using **LLMs**, and expose actionable insights to frontend systems.
- Designed and implemented event-driven, scheduled, and batch workflows using **Google Workflows** and **Cloud Scheduler**, enabling scalable processing of large prospect datasets.
- Built API-first backend services using **FastAPI + Uvicorn**, deployed on **Cloud Run**, to ingest data from databases and third-party data brokers and enrich it via LLM-powered summarization.
- Integrated **Vertex AI** endpoints to manage prompt execution, model selection, and inference governance across multiple application features.
- Implemented dynamic model routing logic to intelligently switch between **Gemini Flash** and **Gemini Pro** based on latency, complexity, and cost requirements.
- Designed secure service-to-service communication using **GCP IAM** service accounts, minimizing blast radius while integrating external data sources.
- Architected **scalable batch and streaming data pipelines** using **Kafka/Pub Sub, Spark/Dataflow, and Airflow**, ingesting large volumes of structured and unstructured data for analytics and AI workflows.
- Designed and optimized **Python and SQL-based ETL/ELT pipelines and dimensional data models**, enabling standardized business metrics and analytics-ready datasets.
- Built **GenAI-powered data pipelines using LLMs and RAG architecture**, integrating vector embeddings and external data sources to generate AI-driven insights on top of a cloud lake house platform.
- Utilized **Cloud Logging** and **Cloud Monitoring** to establish observability across AI pipelines, tracking execution latency, failures, and model behavior.
- Implemented retry, timeout, and fallback mechanisms within **Google Workflows** to ensure resilience and reliability of **AI-driven** processes.
- Enabled horizontal scalability and high availability by adopting serverless-first architecture with **Cloud Run** auto-scaling.
- Collaborated with frontend teams to deliver structured, **AI-generated** insight APIs, reducing frontend complexity and improving UX consistency.

JP Morgan Chase & Co, Albany, NY

Sep 2022- Jun 2025

Role: Senior Cloud Data Engineer

Roles & Responsibilities:

- Involved with project team members to deliver data models to meet data requirements and create all documentation and deliverables in accordance with enterprise guidelines and standards. Work on Metadata transfer amongst various proprietary systems.
- Partnered with ML Engineers and Data Scientists to build AI-driven solutions including chatbots, RAG pipelines, and embedding workflows leveraging **AWS Gen AI** services such as **Amazon Bedrock, Amazon Titan models, Amazon SageMaker JumpStart, and SageMaker Pipelines**.

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- Designed and implemented **Retrieval-Augmented Generation (RAG)** architecture using Amazon **Kendra** for semantic search and **Bedrock-hosted LLMs** for response generation.
- Designed and maintained scalable data pipelines supporting **Retrieval-Augmented Generation (RAG)** architecture to power intelligent conversational agents with real-time contextual knowledge.
- Developed chatbot frameworks integrated with **Amazon Lex, Comprehend**, and **Bedrock** to support intelligent, multilingual, and context-aware conversations.
- Created vector-based feature stores and embedding pipelines using **SageMaker + FAISS**, enabling efficient semantic search across structured and unstructured data.
- Designed and developed data pipeline for **Gen AI (Generative Artificial Intelligence) & machine learning platform**.
- **Led end-to-end discovery and reverse engineering** of complex legacy SQL Server systems (500+ tables, 1500+ stored procedures), **accelerating modernization readiness** by uncovering critical data flows, dependencies, and hidden business logic.
- **Reverse-engineered SSIS packages and Microsoft Access applications, streamlining migration planning** by documenting cross-system dependencies, transformation logic, and undocumented workflows.
- Designed and developed data pipeline using **Docker, Kubernetes, Logstash, Hadoop** and **AWS Stack**.
- Automated daily **ETL** process using **Apache Airflow** and **Crontab**.
- Created **Terraform resources** and **modules** to deploy AWS services like **S3 buckets, Lambda Functions, Glue Jobs, EMR clusters, Step Functions, IAM roles and Policies, Athena** and **QuickSight**.
- **Engineered a centralized metadata repository** consolidating schema details, lineage mappings, and object dependencies, **enabling searchable and auditable insights** for downstream analytics and modernization teams.
- **Created data dictionaries, lineage diagrams, and dependency maps** that **improved system transparency and reduced onboarding time** for analysts and architects supporting migration initiatives.
- Created **YML** files to use **Docker** images, configured to **GitLab CI/CD** pipeline with **Terraform**.
- Created a mechanism (Data pipeline) to collect data from application running in **Docker** and **Kubernetes** to store data in **Hadoop** and **AWS**.
- Moved large amount of data from **AWS S3** buckets to **AWS Redshift** using **Glue** and **EMR**.
- Analysed large and critical datasets using **EMR, Glue** and **Spark**.
- Developed **distributed data pipelines using Spark, Kafka, and Airflow** to process high-volume transactional and behavioral datasets for enterprise analytics platforms.
- Implemented **robust ETL frameworks and data modeling strategies**, transforming raw operational data into curated OLAP-ready datasets used by BI tools such as Tableau.
- Established **data governance, quality monitoring, and observability practices** to ensure reliability, compliance, and availability of mission-critical data pipelines.
- Experienced in writing live **Real-time Processing** and core jobs using **Spark Streaming** with **Kafka** as a data pipe-line system.
- Coordinated daily, weekly, bi-weekly, monthly, and quarterly oversight monitoring activities using **AWS QuickSight**.
- Developed ETL pipelines in and out of data warehouse using tools like **Python** and **AWS Glue**.
- Implemented Spark using **PySpark** and **Spark SQL** for faster testing and processing of data.

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- Consumed the data from **Kafka** using **Apache spark**.
- Used **Spark** and **Spark-SQL** to read the parquet data and create the tables in hive using the **Scala API**.
- Building models in spark using python in **Databricks** and **SageMaker Notebooks**.
- Worked on migrating **SQL** scripts from **Redshift** and **Athena**.
- Experience in building **CICD** pipelines for testing and production environments using **Terraform**.
- Proficient with container systems like **Docker** and container orchestration like **EC2 Container Service, Kubernetes**, worked with **Terraform**.
- **Partnered with SMEs and business stakeholders** to validate extracted logic and document implicit business rules, **ensuring alignment between legacy operations and target-state architecture**.
- **Delivered interactive Power BI and Excel dashboards** visualizing metadata, lineage, and dependency insights, **enhancing traceability and expediting MassDOT technical review cycles**.
- Worked on POC's with **Apache Spark** using **Scala** to implement **spark** in project.
- Used **AWS EMR** to **transform** and move large amounts of data into and out of other **AWS** data stores and databases, such as **Amazon Simple Storage Service (Amazon S3)** and **Amazon DynamoDB**.
- Done some POC on new technologies. Followed agile methodology. Suggested and implemented new technologies.

RCG Atlanta, GA

Jan 2018 – Aug 2022

Role: Senior Data Engineer

Roles & Responsibilities:

- Designed, developed and deployed **DataLake** to store Realtime and batch data from applications and databases using **Confluent Kafka** and **AWS S3**.
- Designed, developed and deployed **Datawarehouse** using **AWS Redshift** following best practices and optimized the queries performance and storing the tables.
- Used **AWS Fargate** and **ECS** for scalable deployment of containerized end points inference services built using **Fast API** and Python.
- Integrated **IAM, API Gateway**, and **CloudWatch** with **Machine Learning Models** to ensure secure, auditable, and highly available model access patterns.
- Designed and implemented **scalable ELT pipelines in Snowflake** using Snow pipe, Streams, and Tasks to enable near real-time ingestion of structured and semi-structured data.
- Optimized query performance in Snowflake by **leveraging clustering, micro-partition pruning, and materialized views**, reducing query latency for analytics and reporting by 40%.
- Built **secure and governed data models in Snowflake** by implementing RBAC, data masking, and role-based access policies aligned with compliance requirements (HIPAA/PII).
- Developed advanced data transformations and **machine learning feature pipelines using Snowpark (Python)**, enabling seamless integration of ML-ready datasets for downstream analytics.
- Collaborated with cross-functional teams to deliver **Lakehouse architecture solutions** integrating Snowflake with Databricks, Apache Iceberg, and cloud storage, ensuring scalability and cost efficiency.

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- Developed self-service pipelines for internal data teams using **SNS, Lambda, Glue, EMR, and Athena**, enabling data readiness for **Machine Learning** and **Data Science** workloads.
- Designed, developed and deployed batch pipelines using **SNS, Lambda, StepFunction, Glue, SageMaker, and EMR**.
- Used **QlikView, Kibana, Tableau** and **Apache superset** for visualization.
- Build Self-service data pipelines using **AWS Services** like, **SNS, StepFunction, Lambda, Glue, EMR, EC2, Athena, SageMaker, QuickSight, Redshift** etc.
- Designed and maintained **data warehouse and data mart models** in AWS Redshift and Teradata to support scalable reporting and analytics.
- Built **ETL pipelines and workflows** integrating Teradata utilities, AWS Glue, and Spark, ensuring consistency, accuracy, and SLA adherence.
- Extensively optimized **Teradata queries** and scripts, tuning joins, indexes, and partitions to handle billions of rows efficiently.
- Analysed large-scale **healthcare datasets (claims, eligibility, EMR/EHR)** within Hadoop and Spark, enabling compliance reporting and predictive analytics for healthcare clients.
- Ensured compliance with **HIPAA and other regulatory standards**, embedding security and access controls in pipelines and DWH structures.
- Performed **data validation, profiling, and quality checks** to reduce downstream errors in analytics and reporting.
- Interpreted and presented **data findings** to non-technical stakeholders (healthcare compliance, audit, operations) and technical teams (engineering, architecture).
- Designed **database architecture** and developed self-service **data platform** in **AWS Cloud**.
- Designed and developed **data pipelines**. Automated based on the requirement.
- Created **Terraform resources and modules** to deploy AWS services like **S3 buckets, Lambda Functions, Glue Jobs, EMR clusters, Step Functions, IAM roles and Policies, Athena and QuickSight**.
- Developed complex data pipelines using **AWS resource S3, EC2, Lambda, Glue, EMR, Step Function and RDS**.
- Developed Topic extraction (**Topic Modelling**) model using **LDA** and other customized **NLP** techniques.
- Developed and automated **data normalizing** model using Python libraries.
- Developed process to generate missing data from source files.
- Designed and developed **data cooking model** to normalize and transform raw data.
- Gathering requirements from SME, Project Manager and Business analysis.
- Established **AWS Bastion host** for secure data transaction between application and database team.
- Represented team in few summits.
- Created **POC on Maria DB Audit Plugin**, identified few performance issues then escalated to **Maria DB research and Develop team**.

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S&P Global Charlottesville, VA

Feb 2016- Dec 2017

Role: Data Analyst

Roles & Responsibilities:

- Designed, developed and deployed **DataLake, DataMart, Datawarehouse** in **Google Cloud Platform**.
- Designed, developed and deployed batch and Realtime data pipelines using **GCP services**.
- Extensive experience on Google cloud services like **Google Cloud Storage, Cloud Function, DataFlow, BigQuery, Cloud Composer (Airflow), DataProc, Compute Engine, Cloud Run** etc.
- Designed and developed and deployed Realtime streaming pipelines using **Apache Kafka** and **Confluent Kafka** to ingest Realtime and batch data from **Relational Databases, DataLakes** and **IOT applications data**.
- Designed and developed and deployed various batch data pipelines using **Airflow**.
- Developed various dags, plugins and custom hooks in **Airflow**.
- Designed, developed and deployed streaming pipelines using **PubSub** and **Dataflow**.
- Migrated On-prem **Hadoop ecosystem** DataLake, Datawarehouse and Data Pipelines to **Google cloud Platform**.
- Created Realtime consumers to read data from **Kafka** and load data into **DataLake (Cloud Storage)** and transformed data using **Spark** in **DataProc** cluster.
- Designed and developed streaming data pipelines using **Apache Kafka, Flume, Hive** and **Pig**.
- Built complex ETL process and stored process data in **Hive** and **HDFS**.
- Transformed raw text data from blogs and social networking sites provided from third party vendors for **sentimental analysis, information extraction** and **information retrieval**.
- Performed data profiling to learn the behaviour of various features such as traffic pattern, location, time, Date and Time etc.
- Good understanding on machine learning techniques like **supervised** and **unsupervised learning**.
- Developing database which contains **Tables, Stored Procedures, Functions, Views, Triggers** and **Indexes** in **SQL SERVER** and connecting to existing **CMS system**.
- Created a POC on **image classification model**.
- Understating sales from past few years, using **Machine learning techniques**.
- Extracted data from third party application and conducted data preprocessing and data mining using **R** and **Python**.

IBM, Hyd IN.

Jul 2014- Dec 2015

Role: Data Analyst

Roles & Responsibilities:

- Participated in analysis, design, development, testing, and implementation of various financial Systems using **Oracle, Developer** and **PL/SQL**.
- This system consists of the various functional modules.

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- Define database structure, mapping and transformation logic. Creation of External Table scripts for loading the data from source for **ETL (Extracting Transforming and Loading) Jobs**.
- Wrote **UNIX Shell** Scripts to run database jobs on server side.
- Developed new and modified **existing packages, Database triggers, stored procedure and other code modules using PL/SQL** in support of business requirements.
- Worked with various functional experts to implement their functional knowledge into business rules in turn as working code modules like procedures and functions.

TOOLS AND TECHNOLOGIES:

Languages	Shell, SQL, Python, PySpark and R
Generative AI & LLMs	LLMs: Amazon Titan, GPT-4 / GPT-3.5 (OpenAI), Claude (Anthropic), Gemini (Google), Llama 2 / 3 (Meta), Falcon, Mixtral, DeepSeek, Cohere, BLOOM
Embedding & Vector DBs	SentenceTransformers, Hugging Face Embeddings, FAISS, ChromaDB, Pinecone, Weaviate, Milvus, OpenSearch
RAG & Agentic Frameworks	LangChain, Llama Index, Haystack, CrewAI, AutoGen, PromptLayer
Packages	NumPy, Pandas, PyOd, Spacy, Matplotlib, Seaborn, Bokeh, Beautiful Soup, Selenium, Scikit-learn, TensorFlow, PyTorch, Lime, H2O, Scikit-image, Pillow, Psycopg, SQLAlchemy , Flask and more.
Container technologies	AWS EKS, Docker and Kubernetes.
Cloud Platforms	AWS and GCP.

EDUCATION:

Master of Computer Applications

Osmania University, Telangana, India

Jun 2014

Bachelor of Science B.sc (Mathematics. Physics. Chemistry)

Kakatiya University, Telangana, India

Jun 2010